

United States Census 2030

2030 Census Systems Engineering Management Plan (SEMP)

Approval Signatures

This **2030 Census Systems Engineering Management Plan (SEMP) V1.1** has been reviewed and approved for use.

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Version History

Version	Date	Description
V1.0	10/11/2023	Baseline version sent for signatures.
V1.1	07/18/2024	This aligns the 2030 Census SEMP with the recently baselined 2030 Census Integration and Implementation Plan (IIP), including the Program Level Test Checkpoint (PLTC) as part of the 2030 Census Program Readiness Reviews and Milestones graphic.

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1 Introduction

The 2030 Census Systems Engineering Management Plan (SEMP) outlines the technical and managerial processes that govern system engineering management activities. This plan provides an interdisciplinary approach encompassing the entire technical development effort to evolve and verify an integrated set of systems, people, processes, and solutions that satisfy customer needs throughout the 2030 Census Program's life cycle.

A notable aspect of the 2030 Census Program is the intersection of numerous systems, each with their own set of internal and external users. These complex interactions underscore the importance of adhering to proper standards and guidelines, as outlined in this document. The SEMP is designed to ensure consistency across management and technical activities supporting the 2030 Census.

The SEMP covers various areas such as requirements management and traceability, engineering and architecture, system design and development, technical integration, and other components necessary during the 2030 Census Program's life cycle. Functioning as a high-level guide, the SEMP aims to direct an environment and conditions in which a defined set of goals or objectives can be achieved in a controlled manner by a team of people. It incorporates recent advancements in Software Development Life Cycle (SDLC) and engineering processes, enabling adaptable and efficient management, architecture, development, and system integration tailored to the needs of the 2030 Census Program.

1.1 Intended Audience

This document is intended for all those involved in the Architecture, Engineering, Program Management, and Information Technology (IT) solution development for the 2030 Census, examples of which are listed below:

- Decennial Systems Engineering (DSE) and Technology Integration (TI)
- Decennial Census Management Division (DCMD)
- Decennial Census purpose-built systems
- Enterprise Initiatives Business Ecosystem (BE)
- Office of the Chief Information Officer (OCIO)
- Decennial Contracts Execution Office (DCEO) (contractor-delivered solutions)
- Third-party contracted solutions

1.2 Document Maintenance

The DSE, TI, and Security teams in the Decennial Information Technology Division (DITD) are responsible for creating and maintaining the SEMP. This document is reviewed and updated periodically to verify its continued relevance and accuracy.

1.3 2030 Census Program

Mission and Vision: The 2030 Census mission is to conduct a census of population and housing and deliver the data to the President, the States, and the American people. The program's vision is to conduct an efficient, effective, and quality census that counts the people of our nation, once, only once, and in the right place.

Decennial Census Background: While the Census Bureau administers many censuses and surveys of the U.S. population, commerce, and governments, including the future 2030 Census, the decennial census of population and housing, conducted once every 10 years, is the only one mandated by the United States Constitution. The decennial census is the country's oldest, most comprehensive source of information about the population. The decennial census provides comprehensive statistical information for the United States and U.S. Territories, including demographic and housing characteristics.

The U.S. government uses the data generated by the census to apportion congressional seats, monitor compliance with civil rights and other legislation, and forecast the number of people eligible for government benefits.

Tribal, state, and local governments use census figures to distribute funds for education and other social programs; determine congressional, state legislative, and voting district boundaries; and predict disaster relief needs.

Island Area governments use decennial census data to access federal funding opportunities, to better understand the characteristics of a changing population, such as age and household composition, and to understand and address emerging needs, such as access to employment, health insurance, and internet access. In addition, decennial census data for the Island Areas of the Pacific and the U.S. Virgin Islands are the sole data source for federal agencies.

Business relies on the data for product development, marketing, demand forecasting, and determination of optimum location sites.

2030 Census Program Life Cycle: During the planning and preparations for the 2030 Census, the Census Bureau will continue to strive to improve the accuracy of census coverage compared with previous censuses, reduce operational risk, and contain costs, while ensuring the

robustness of the design to ensure a quality count. Planning for the 2030 Census began in 2019. The 2030 Census Program is being conducted in four phases, beginning with Early Planning (FY19-FY21), followed by Design Selection (FY22-FY24), Development and Integration (FY25-FY29), and Peak Production and Close-Out (FY29-FY33).

1.4 Systems Engineering Management Plan Outline

The SEMP is organized into three main sections. The SEMP is not all-inclusive. Where applicable, it references documents that provide specific and detailed guidance (e.g., Integration and Implementation Plans (IIPs), System Design Documents).

Section 2 - System Engineering Life Cycle

The System Engineering Life Cycle section outlines a systematic approach to achieving the activities that occur during the various phases of the System Engineering life cycle. This section is organized based on a modified version of the Engineering V-Model, as it depicts a more iterative, agile approach than the traditional model implies. By adhering to this model, the SEMP can better explain key concepts from conceptualization to deployment – including Readiness Reviews (RRs), supporting processes, and artifacts. It also describes the differences in roles and responsibilities for all levels of activities (program and project) and their associated RRs.

Section 3 - Systems Engineering Management

The System Engineering Management section identifies the management strategies and methodologies that guide successful systems engineering. It identifies how the Scaled Agile Framework (SAFe) is used to manage the System Engineering life cycle and defines and details the role of operational delivery management. It also describes the governance, security, decision-making processes, schedules, and planning that collectively manage the System Engineering life cycle.

Section 4 - Engineering and Specialty Integration

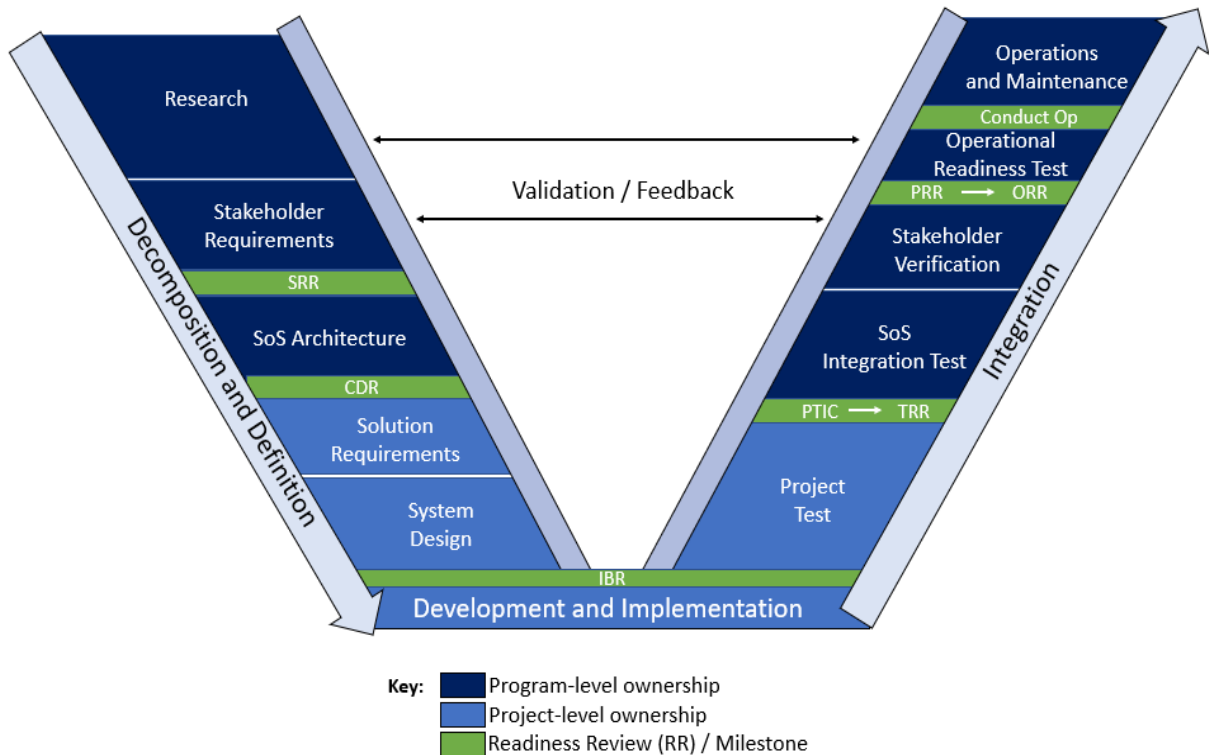
This section provides details regarding engineering specialties that are integrated with the System Engineering life cycle. Specifically, this section discusses security protocols and internal and external audit assessments.

2 System Engineering Life Cycle

The 2030 Census Program uses the V-Model approach to navigate the System Engineering life cycle. This approach combines the principles of Agile methodology with the structured and

systematic approach of the traditional V-Model. This model is well-suited for the 2030 Census Program, as it provides a balanced framework for managing complexities, facilitating iterative development, and ensuring seamless integration across varying project elements. The model is designed to be flexible. It accommodates various development methodologies including Agile, Waterfall, and Incremental. These methodologies align to the Census Bureau's Enterprise Systems Development Life Cycle (eSDLC).

Figure 1: V-Model Life Cycle



The model consists of three key stages, as shown in Figure 1:

Decomposition and Definition: This stage is depicted on the left side of the V-Model. During this stage, high-level requirements and objectives are decomposed into smaller, more manageable components. These components serve as the basis for solution requirements.

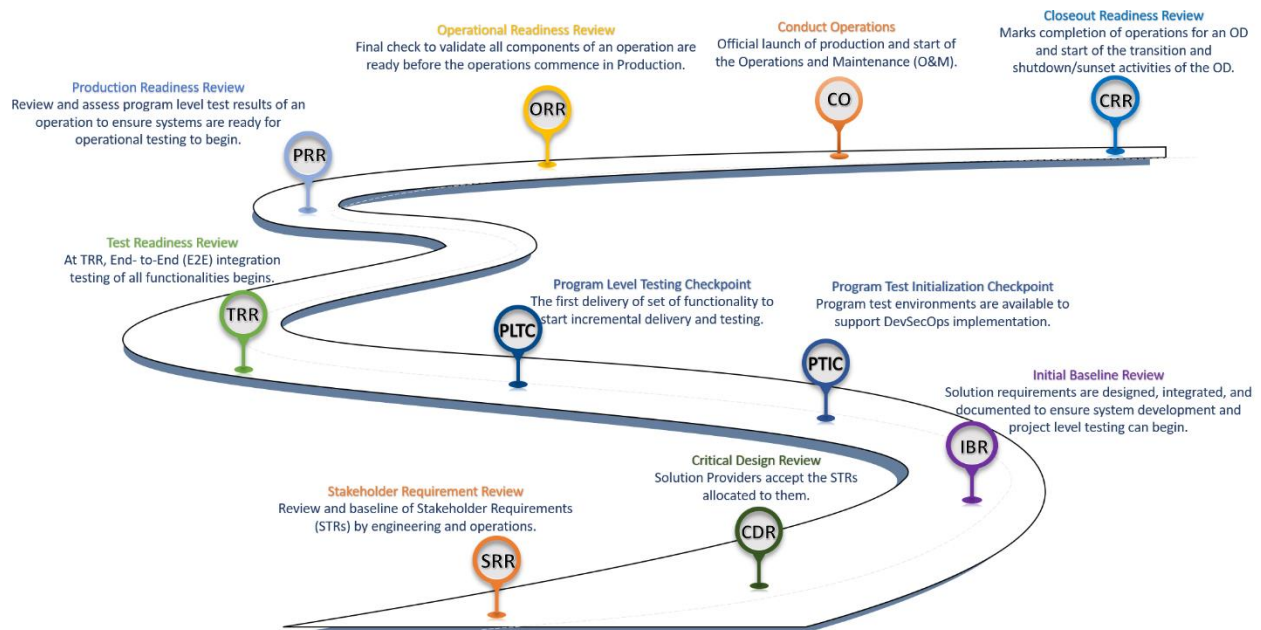
Development and Implementation: This stage is represented by the base of the V-Model. At this stage, solution providers work on building and testing individual components. The V-Model offers flexibility and adaptability, acknowledging the dynamic nature of software development and recognizes that teams may adopt either Agile, Waterfall, or Incremental development practices based on project suitability.

Integration: The final stage, depicted on the right side of the V-Model, is Integration. Throughout this stage, components developed during the previous stage are integrated and tested as a whole, ensuring they function cohesively and meet the overall 2030 Census Program requirements.

The V-Model encompasses a combination of 2030 Census program-owned and project-owned activities. Program-owned activities are concerned with defining high-level goals, outcomes, the 2030 Census Program vision, and inter-project coordination. Project-owned activities focus on specific deliverables and objectives of a single solution within the 2030 Census Program.

Throughout the execution of the life cycle, the 2030 Census Program conducts RRs and milestone ceremonies. RRs and milestones are structured events that assess the readiness of the 2030 Census Program to proceed to subsequent activities. Figure 2, below, outlines RRs and milestones associated with various activities conducted throughout the V-Model. The RRs and milestones are described in further detail throughout this document.

Figure 2: Program RRs and Milestones

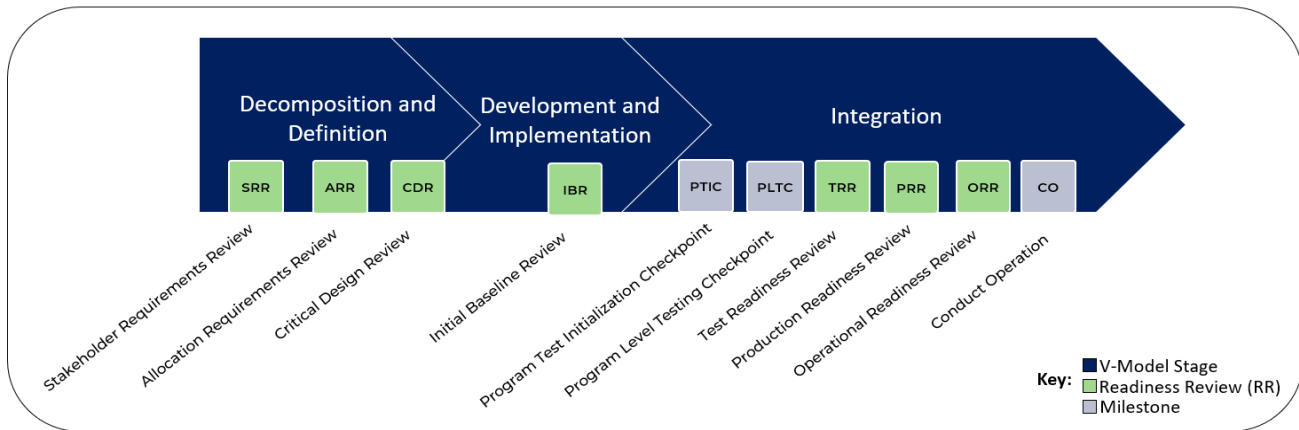


The primary objectives of the RR milestones are to:

- Evaluate readiness to determine whether the 2030 Census Program is prepared and capable of achieving its objectives.
- Identify risks and issues that could prevent the 2030 Census Program from achieving its objectives.
- Provide recommendations for improving the 2030 Census Program's readiness and addressing identified risks and issues.

- Enforce compliance with regulations, policies, and standards.

Figure 3: 2030 Census Program RRs and Milestones



The sequence of RR milestones is illustrated in Figure 3 above.

At Stakeholder Requirements Review (SRR), the business presents stakeholder requirements to the 2030 Census Program. This process continues until all requirements are shared, baselined, and allocated to solution providers. As SRRs occur for each operation, and as the 2030 Census Program continues to allocate stakeholder requirements, the Solution Architecture Team, in parallel, builds the System of Systems (SoS) high-level architecture designs. This work is ongoing until all SRRs are complete, allowing the 2030 Census Program to proceed with the Critical Design Review (CDR). The CDR is a singular event in which the SoS high-level architecture and supporting artifacts are reviewed and baselined. The solution providers then have all the necessary inputs to identify and refine solution requirements.

As solution providers continue defining solution requirements, iterative Initial Baseline Reviews (IBRs) occur. Detailed requirements and associated artifacts can then be reviewed, indicating development and implementation can begin. Key milestones after IBR are the Program Test Initialization Checkpoints (PTIC), Program Level Testing Checkpoint (PLTC), and Test Readiness Reviews (TRR) that assess the 2030 Census Program's readiness to start SoS integrated program-level testing.

Once all program-level testing is complete and the SoS has been validated, Production Readiness Review (PRR) is conducted to allow operational teams to conduct final validation of operational aspects of the SoS. Successful operational readiness testing is the entry criteria for Operational Readiness Review (ORR), indicating readiness to Conduct Operations (CO).

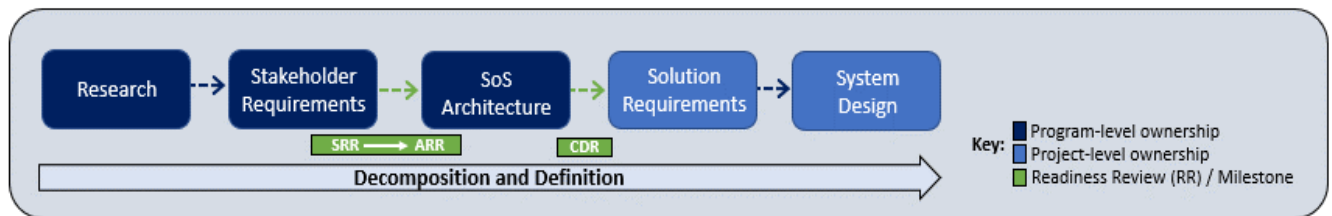
These milestones are mentioned throughout the SEMP, where applicable, as they relate to the System Engineering life cycle.

2.1 Decomposition and Definition

The Decomposition and Definition stage of the V-Model, shown in Figure 4, sets the foundation for the 2030 Census Program's success. By conducting thorough research on customer needs and objectives and breaking down requirements into well-defined measurable capabilities, the 2030 Census Program ensures a clear understanding of the scope and architectural needs of the SoS.

This stage consists of five activities: Research, Stakeholder Requirements, SoS Architecture, Solution Requirements, and System Design. Responsibilities at this stage are divided between the 2030 Census Program and solution providers.

Figure 4: V-Model – Activities within the Decomposition and Definition Stage



2.1.1 Key Activities

2.1.1.1 Research

Research is a key step in decomposing and defining the 2030 Census technical solution. The 2030 Census Program takes a three-phased approach to engineering research: Understand, Conceive, and Evaluate.

Understand: This phase focuses on comprehending the 2030 Census Program's context, the problems to be solved, and the needs of the stakeholders. In some cases, problems are clearly understood and/or previously identified by lessons learned. For these problems, the approach calls for minimal or moderate collaboration with subject matter experts (SMEs) to define requirements. In other cases, little is known about the problem itself or potential remediations. The objective is to gather insights that serve as the foundation for subsequent decision-making and solution development.

Conceive: This step involves brainstorming and generating potential resolutions to the problems outlined. In this step, different ideas, concepts, and approaches are explored to envision how the 2030 Census Program can address identified needs and challenges.

Evaluate: During this phase, the 2030 Census Program critically assesses the potential solutions generated in the “Conceive” step. This assessment involves feasibility, practicality, alignment with 2030 Census Program objectives, and the ability to meet customer needs. The 2030 Census Program uses data-driven methods which include a set of defined, weighted criteria applied to each candidate solution, allowing for accurate comparison. DITD assesses five foundational parameters including total cost of ownership, maturity, subsequent applicability to the Census Bureau enterprise, scalability, and security. These results, assessments, and recommendations then progress through the appropriate IT governance implementation. The goal is to narrow down the list of potential solutions that are most promising and aligned with 2030 Census Program goals.

2.1.1.2 Stakeholder Requirements

The 2030 Census Program manages four levels of requirements: **Mission, Activity, Stakeholder, and Solution**. Solution requirements are described in section 2.1.1.4 Solution Requirements.

Mission requirements are externally facing requirements that establish why the 2030 Census Program is required. **Activity** requirements define the actions necessary to achieve the outcomes of one or more mission requirements. Both are considered high-level requirements that lay out the 2030 Census Program’s overall mission and operational outcomes that drive the development of stakeholder requirements.

Stakeholder requirements are mid-level requirements which represent the needs of the 2030 Census Program’s internal stakeholders to achieve Mission and Activity requirements. Stakeholder requirements place the stakeholders and users at the center of focus and describe what the 2030 Census Program must produce or provide. Business stakeholders are responsible for defining and baselining stakeholder requirements. Once developed, stakeholder requirements are analyzed to determine if the solution provider(s) should be an IT system, an operational solution, or both. For additional details on stakeholder requirements, refer to the 2030 Census Requirements Management Plan (RMP) listed in Appendix C.

Solution requirements are developed for both IT and operational solutions. They are the most detailed program requirements providing solution providers with manageable and logical information needed to develop solutions.

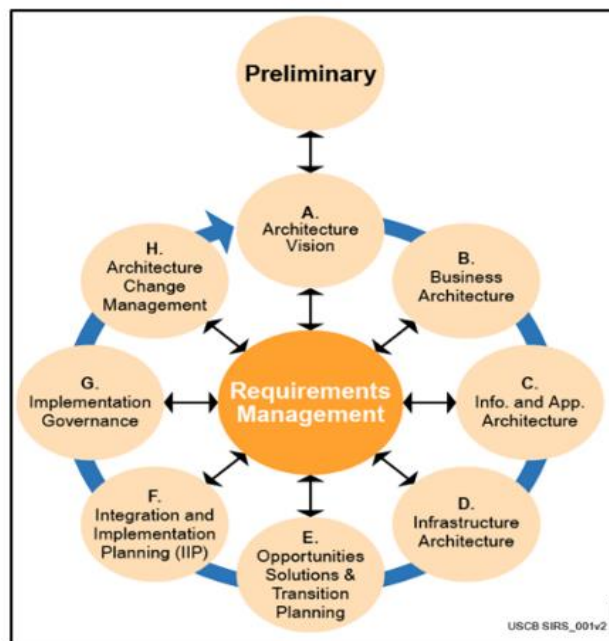
SRR is the first milestone within the V-Model. The objective of SRR is for the business to deliver stakeholder requirements and supporting documents such as a Business Process Model (BPM) to DSE and TI. SRRs are iterative events which allow the SoS architecture to begin formulating designs while new stakeholder requirements are received, and baseline and allocation activities are continuously conducted. The Solution Architecture Team allocates the Stakeholder-level IT

requirements to solution providers throughout SRR and delivers final allocated stakeholder requirements to prepare the 2030 Census Program to enter the next milestone: CDR. By conducting SRRs, the program ensures all solution provider objectives and expectations are clear. This clarity reduces any ambiguity and minimizes the risk of misinterpretation, leading to a more consistent, successful, and aligned SoS design. It is key to note that allocation activities and design of the SoS architecture may occur in parallel.

2.1.1.3 SoS Architecture

The 2030 Census Program designs and implements the SoS architecture to enable business needs and objectives and to align with the 2030 Census IT Strategy and Roadmap. The architecture is documented in the 2030 Census SoS Solution Architecture document. The process of developing the architecture is based on "The Open Group Architectural Framework" (TOGAF), a cyclical Architecture Development Methodology (ADM).

Figure 5: TOGAF-based Architecture Development Method



During the **Preliminary** phase shown in Figure 5, the Solution Architecture Team reviews various artifacts and documents to understand the 2030 Census Program's vision, goals, and objectives. This review includes identifying frameworks, methods, and processes that are used to develop the architecture, defining the organizational model for the architecture, establishing the detailed process and resources for Architecture Governance, and selecting and implementing tools that support the architecture. Activities in each of the steps usually overlap, and multiple iterations of the steps may be implemented as additional elements of the architecture evolve through testing, change requests, or lessons learned.

Step A - The Architecture Vision, provides a high-level description of key approaches to be used within the architecture and defines the Architecture Principles captured in the 2030 Census Architecture Vision document.

Step B - The Business Architecture, describes how the solution needs to operate to achieve the business goals and respond to the strategic drivers set out in the Architecture Vision in a way that addresses stakeholder concerns. The Program Architecture is documented in both BPMs and Integrated Operations Diagrams (IODs). Stakeholder requirements, BPMs, and IODs are baselined at SRR.

Step C - The Information and Application Architectures, describes the data repositories and high-level data structures to be used as well as the partitioning of the highest level SoS into systems. As part of this step, stakeholder requirements are allocated to the systems. Communication diagrams, referred to as Workflow Diagrams (WFDs), are also created during this step. The purpose of a WFD is to provide a clear and comprehensive overview of how systems communicate with each other during an operation. It outlines all participating systems, interfaces, processes, interactions, and users involved. Because WFDs include lower-level interfaces, draft WFDs may be created as the application architecture is being developed. The WFDs (and the interface catalog) are not baselined until after solution providers have begun development and started to define the interfaces. WFDs and the interface catalog are baselined at IBR.

Step D - The Infrastructure Architecture, describes, at a high level, the physical architecture used in the SoS, which is predominantly physical locations of facilities, high-level network diagrams, and selections for cloud/on-prem locations at a high level, also known as the “what goes where diagram,” for each system. It is after this step that solution providers are identified or notified. After IT solution providers review and accept the requirements, they begin the process of working with stakeholders to further decompose/clarify the requirements, operations, and business processes that have been allocated to them.

Step E - Opportunities, Solutions, and Transition Planning, is the creation of the 2030 Census Architecture Transition Plan that describes the architectures for the 2026 Census Test, 2028 Dress Rehearsal, and the 2030 Census. The transition plan helps drive a development and integration approach in which the solutions are built incrementally as they progress to the final solution. The incremental development approach drives the creation of the detailed IIP.

Step F - Integration and Implementation Planning, which guides the development and integration of the individual systems.

Steps G - Implementation Governance, and **H, Architecture Change Management**, then provide governance, management, and control of these architecture artifacts.

The SoS architecture provides the high-level vision of the entire SoS. It illustrates how different subsystems or components interact and contribute to the overall 2030 Census Program's goals. The CDR is the milestone in which SoS Architecture artifacts, such as the Application Architecture Diagram, are communicated and baselined. The solution providers can then begin solution requirement elicitation, acquisition planning, if necessary, and development.

2.1.1.4 Solution Requirements

The requirements analysis phase of the eSDLC includes gathering, analyzing, and documenting solution requirements. Solution requirements are the lowest level of requirements that provide manageable and logical details required to develop units of functionality and, ultimately, a solution.

Solution requirements can either be fulfilled by an IT solution, an operational solution, or both. Solution providers, in collaboration with SMEs, are responsible for identifying and documenting their respective solution requirements using the SoS architecture and stakeholder requirements as inputs.

The development of solution requirements embodies an iterative and incremental approach. Through detailed planning, solution providers, in collaboration with SMEs, engage in the process of refining and shaping requirements in response to changing customer needs as well as the identification and allocation of new stakeholder requirements. This iterative practice demonstrates how solution providers break down complex requirements into manageable units of work (or user stories), fostering incremental development. As this iterative practice progresses, new insights are seamlessly integrated, ensuring that the solutions remain aligned with 2030 Census Program objectives.

For additional details on solution requirements, including rules that must be adhered to when documenting solution requirements and the level of traceability expected from solutions requirements, refer to the 2030 Census RMP listed in Appendix C.

2.1.1.5 System Design

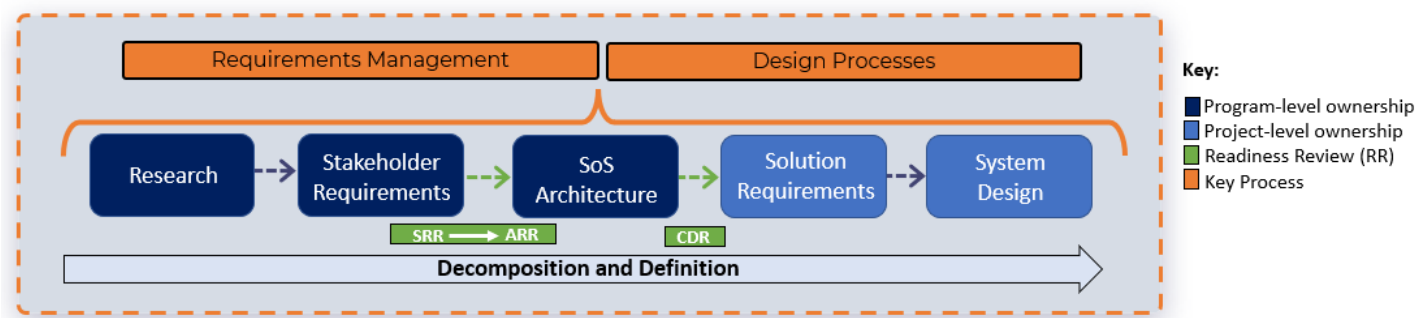
System design refers to the process of creating an architectural plan for a solution that meets specified solution requirements. Typically, system design involves breaking a solution down into modules and defining relationships and interfaces between them. The identification of modules supports solution providers by further decomposing them into components that define the internal structure of each model, including any data structure, algorithms, and interfaces within each component.

System design is a critical step that bridges the gap between requirements and implementation. It ensures that the system's architecture is well-structured, components are well-defined, and the solution's design aligns with intended and required functionality. The output of system design serves as the basis for the next stage in the engineering life cycle: Development and Implementation (see section 2.2).

2.1.2 Key Processes Applied – Decomposition and Definition

Decomposition and Definition activities include two key processes as shown in Figure 6. The use of standardized processes contributes to the success of the 2030 Census Program by providing a consistent and predictable framework that ensures activities are executed in a reliable manner across the program. As mentioned previously, this stage sets the foundation for the 2030 Census Program's success and requires that robust, structured procedures be in place.

Figure 6: Decomposition and Definition Key Processes



2.1.2.1 Requirements Management

Requirements Management is a fundamental and critical aspect of System Engineering that serves several purposes, including setting clear expectations, defining scope, establishing alignment with stakeholder needs, and providing comprehensible reporting and traceability.

The Requirements Management Team (RMT) oversees Requirements Management and is responsible for facilitating the collection, analysis, and on-going maintenance of requirements and maintaining the RMP. The RMP outlines the requirements management process from elicitation to validation and verification and provides a framework for how requirements are collected, analyzed, documented, and maintained throughout the 2030 Census Program life cycle.

2.1.2.2 IT Solution Design Processes

The eSDLC defines the project-level engineering processes, along with the necessary artifacts and reviews, for all 2030 Census solutions. In addition to following the eSDLC process, all

solution providers supporting the 2030 Census must have solution designs reviewed by the DITD Engineering Working Group (EWG). The EWG reviews all aspects of a solution's architecture to ensure the design:

- Meets Architectural standards, especially in relation to cloud smart architecture.
- Satisfies security and infrastructure requirements.
- Maximizes cost efficiency.
- Aligns with 2030 Census business needs.

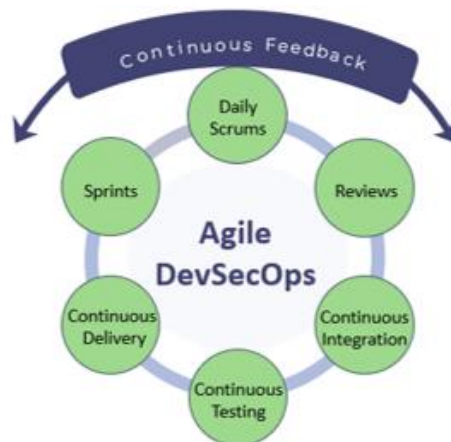
Additionally, the 2030 Census SoS Solutions Architecture document outlines design principles that should be applied, when possible, at the project level.

2.2 Development and Implementation

Development and Implementation activities begin following the IBR milestone. The goal of IBR is to ensure system artifacts, such as solution requirements (features and user stories or equivalent), architecture artifacts, technical specifications, and engineering plans are accepted by participating systems and leadership before beginning development and testing.

Development and Implementation refers to the process of actualizing the system design into a functional solution, which may encompass building the components, integrating them, and ensuring that the system is developed as intended.

Figure 7: Agile DevSecOps



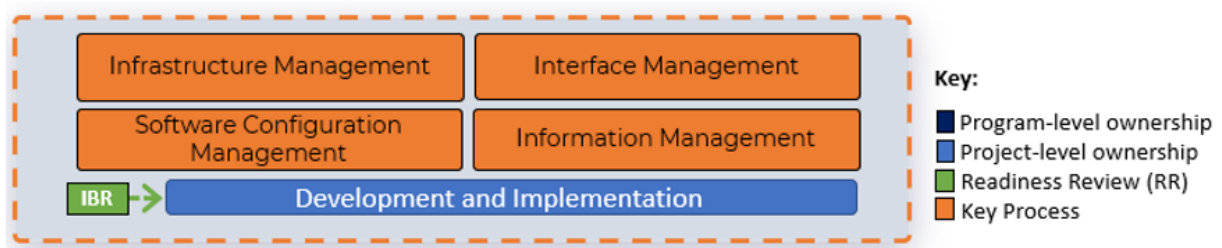
Solution providers can employ different software development processes tailored towards their requirements, integration, and data sensitivity; however, they must comply with 2030 Census standards and guidelines as provided by the DITD. The eSDLC serves as an organized process model for developing and implementing a system or solution.

The 2030 Census Program recommends optimization and modernization of 2030 Census solutions as appropriate, driven by cloud-native pillars such as microservices, containerization, automation, Infrastructure as Code (IaC), and Development Security Operations (DevSecOps) principles. The 2030 Census Program envisions these technologies as the foundation for software development efforts. By leveraging modern methodologies, solutions achieve scalability, continuous improvement, resilience, and efficiencies in deployments. Furthermore, working under a DevSecOps (see Figure 7) approach verifies that security is integrated at every stage of the development life cycle. Teams can then obtain immediate feedback on adherence to coding standards, code quality, and security requirements.

2.2.1 Key Processes Applied – Development and Implementation

Infrastructure Management, Interface Management, Software Configuration Management, and Information Management are all key processes applied for effective development and implementation (see Figure 8). The use of standardized processes and best practices contribute to the success of the 2030 Census Program by fostering cooperation, streamlining processes, and safeguarding the integrity of system data and infrastructures.

Figure 8: Development and Implementation Key Processes



2.2.1.1 Infrastructure Management

The methodology for designing, implementing, and managing the infrastructure for the 2030 Census is described in the 2030 Census SoS Infrastructure and Integration Plan (SIIP). The SIIP provides details on the architecture and integration mechanisms for Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS), as well as other cloud provider managed service components that are used to support the 2030 Census. It outlines an infrastructure architecture that is scalable and capable of handling innovation to meet the demands of the 2030 Census. It details how various infrastructure components are integrated to provide a cohesive infrastructure solution. Lastly, the SIIP outlines how change is managed to verify impacts are understood and evaluated for risk.

2.2.1.2 Software Configuration Management

The Software and Infrastructure Configuration Management Plan (CMP) manages the life cycle of software and infrastructure changes. The 2030 Census Program strongly encourages solution providers to adopt Continuous Integration/Continuous Delivery (CI/CD) practices and use a suite of DevSecOps tools and governance processes. These practices improve the quality of the deliverables, minimize risk, and facilitate an agile and timely response to changes in operational configurations or specifications during production operations.

Configuration and change management are tightly integrated in the 2030 Census Program. They provide a clear escalation path for technical changes that may impact the program (as described in Section 3.7: Change Management).

2.2.1.3 Interface Management

The Interface Management Group (IMG) serves as a technical arm of the 2030 Census Program. The IMG assesses and facilitates the technical maturity of system interfaces. The applicable standard templates and procedures associated with the IMG process are defined in the 2030 Census Interface Document Governance Process, listed in Appendix C.

The 2030 Census Interface Catalog (listed in Appendix C) documents all data flows between Census Bureau systems to be used in support of the 2030 Census and includes intersystem data flows. There are two types of interfaces: Enterprise-identified interfaces and 2030 Census-identified interfaces. An **Enterprise-identified interface** is any 2030 Census applicable interface where the producer or consumer is an Enterprise system (e.g., Enterprise Data Lake (EDL) or Data Ingest and Collection for the Enterprise (DICE)). The Enterprise Interface Catalog is an authoritative document for all referenced interfaces related to the Enterprise. **2030 Census-identified interfaces** are all interfaces outside of the enterprise scope, but still applicable to the 2030 Census Program.

The 2030 Census Solution Architecture Team interacts with IT solution providers as needed to resolve any synchronization issues. If there are any changes that add, modify, or remove interfaces in the 2030 Census Interface Catalog, the 2030 Census Solution Architect or Solution Provider initiates a 2030 Census change request. The 2030 Census Interface Catalog is available for review by stakeholders in the 2030 Census Configuration Management (CM) repository.

2.2.1.4 Information Management

Information Management is a set of processes, policies, and design patterns that confirm information is properly stored, maintained, secured, and accessible only to those with a business need. In the context of the 2030 Census, these processes are designed to facilitate the

use of advanced data science methods and techniques, transactional processing, statistical analysis, and the application of artificial intelligence and machine learning (AI/ML). All 2030 Census solutions must adhere to the Information Management policies, with an emphasis on the Interface and Security policies, described in the Enterprise SEMP.

2030 Census solutions which require use of Personally Identifiable Information (PII), Title 5, Title 13, or Title 26 data work with an assigned Decennial Information Security Engineer, or the Information System Security Officer (ISSO) from the Office of Information Security (OIS) to obtain the necessary security authorizations (defined by OIS) for intended testing and production activities. Additionally, the Information Security Engineer, or ISSO, approves an Authorization to Operate (ATO) for all systems before their release to production. For additional information on security management, see Section 4.1.

In addition to adhering to enterprise security policies related to information management, the 2030 Census embraces the Census Bureau's digital transformation effort, including leveraging advanced data science methods and techniques to unlock insights and enhance efficiency. Key areas of focus are:

- BE initiatives: Using the full potential of Census Bureau resources and data analytics efforts to improve the quality and efficiency of the 2030 Census.
- Strategic leverage of administrative data and advanced record linkage approaches: Expanding the use of previously independent data sets by implementing strategic approaches to link and use administrative data.
- Incorporation of cloud-based analytics platforms: Enabling scalable and cost-effective storage, processing, and analysis of extensive 2030 Census data, which support real-time analytics and decision-making.
- Predictive analytics and modeling: Using large sets of historical and administrative data to predict activities and drive more efficient approaches in 2030 Census operations.
- Applying advanced data science methods and techniques including AI/ML in 2030 Census Program planning, execution and management, security, software development, test, and other stages of the SDLC.
- Attention to cost effective storage and retention approaches including online storage needs for databases and active repositories as well as long-term storage needs for backups and archival.
- Given the data-centric nature of the SoS Architecture, sensitive data is centrally located. While this centralization helps control the dispersion of data, it also provides adversaries a single “target.” As such, significant attention needs to be given to protecting this central repository and avoiding breaches, rather than simply detecting them.

- Impact of updates to administrative data and Frames data: Using administrative data and Frames data as primary sources prompts a need for data management that does not disrupt users of that data. This data requires strategies for data release, quality measurements, and accuracy indicators.
- Usability by 2030 Census: To implement the 2030 Census architecture vision, the use of multiple administrative data sets throughout the decade becomes critical. As such, determining the applicability and quality of those data sets is paramount.
- Extension to Data Management: Traditional software control and management processes must now also apply to data, ensuring consistency and accuracy.

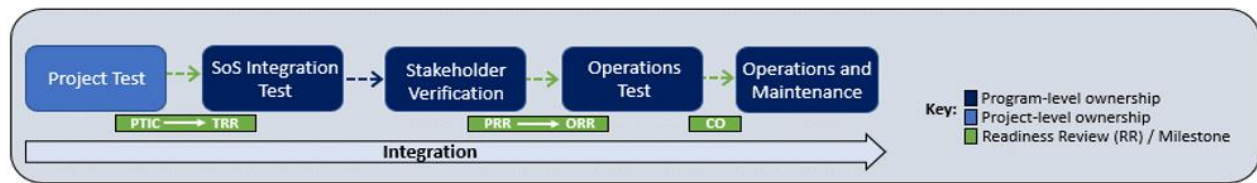
The Information Management efforts for the 2030 Census Program also focus on planning for the integration of applications with EDL, establishing data pipelines, implementing Extract/Transform/Load (ETL) processes, standards-based data storage, and data re-use.

The overarching goal is to leverage modern technologies and data-driven approaches to enhance decision-making and drive overall efficiency and productivity. Preparing for these advanced data-centric challenges requires enhancements to processes and policies to drive governance, standardization, privacy protection, security, communication, and other activities.

2.3 Integration

The Integration stage of the V-Model aims to bring together the individual components or sub-systems developed during the Development and Implementation stage to validate that the integrated solutions work harmoniously as a complete SoS, with data flows behaving as expected from solution A through solution Z. The Integration stage is essential for verifying that the integrated SoS meets specified requirements, interfaces between components are functioning correctly, and any potential issues or inconsistencies are identified and addressed prior to conducting operations in Production.

This stage consists of five activities: Project Test, SoS Integration Test, Stakeholder Verification, Operations Test, and Operations and Maintenance (O&M). Stakeholder responsibilities at this stage are divided between the solution providers and the 2030 Census Program. Solution providers own their respective project test activities, and the 2030 Census Program manages the subsequent integration efforts.

Figure 9: V-Model – Activities within the Integration Stage

2.3.1 Key Activities

2.3.1.1 Project Test

Project testing is the responsibility of the solution providers, and may occur in parallel with Development and Implementation, as solution providers progress iteratively in delivering components of functionality and testing. During the early phases of development, project testing is conducted to demonstrate the feasibility of conceptual approaches, evaluate design risk, identify alternatives, compare and analyze trade-offs, and assess the solution’s satisfaction of requirements. The project-level test scope is comprised of various test efforts including Unit Test, Component Test, Pairwise Test, Application Programming Interface (API)/Component Base Performance Test, and User Acceptance Test. For additional details on each of these test efforts, refer to the Test and Evaluation Management Plan (TEMP), listed in Appendix C.

With the adoption of IaC and a CI/CD pipeline, most project testing can be achieved through automation. Automation provides continuous validation of updates as changes are developed and merged. Continuous validation avoids the need for environment lockdown in cases where developers are working in parallel and reduces the risk of code changes that break existing functionality.

During project test activities, solution providers are also expected to collaborate with infrastructure teams and program-level test (PLT) teams to prepare for upcoming 2030 Census Program test efforts. This milestone is referred to as the PTIC. The PTIC is conducted to certify that teams are prepared to perform kickoff activities required by the PLT efforts, including communication of expected deliverables, provisioning of higher environments, and implementing CI/CD pipelines in said environments. At a minimum, criteria to “exit” PTIC includes a ready-for-use Integrated Test Environment (ITE) and access to the ITE granted to members of the 2030 Census PLT team. For additional details on PTIC, refer to the 2030 Census IIP and the 2030 Census TEMP listed in Appendix C.

Project testing is an ongoing activity that continues throughout the SDLC. CI/CD forms a foundation for modern software development in that it enables project testing to persist concurrently with program testing, seamlessly intertwining both activities through a well-orchestrated pipeline.

2.3.1.2 SoS Integration Test

The primary objectives of SoS Integration test activities are to confirm a solution's functionality, performance, reliability, and validate that the SoS meets specified requirements. Following a successful PTIC and the verification of a preliminary set of solution capabilities during project test, the 2030 Census PLT team conducts the TRR. The objective of TRR is to confirm appropriate 2030 Census Program test objectives, schedules, and scripts have been documented, environments are ready, and solution providers are prepared to participate in integrated program-level testing. This milestone initiates program-level SoS Integration Testing (SIT).

SIT is the process of combining various systems into one overall SoS. A multitude of test efforts are conducted during SIT. These efforts include but are not limited to: End-To-End Integration (EEI) testing, device and browser testing, exception testing, and Performance and Scalability (P&S) testing in an environment(s) mirroring the target infrastructure of Production. SIT focuses on executing tests across system boundaries. Such testing validates the physical and logical interfaces between systems, hardware products, software products, and external interfaces. For more information on program test scopes, such as ownership, support area(s), and timeline, refer to the TEMP and the P and S Test Plan, listed in Appendix C.

Throughout 2030 Census Program test activities, solution providers are expected to support the PLT teams by providing an initial overview of the system, maintaining technical specifications, troubleshooting, and remediating defects.

CI/CD practices are carried forward from project test to program test; however, practical differences may exist, including, but not limited to:

- **Deployment frequency:** Projects tend to deploy changes more frequently to lower environments, where new releases may occur as often as 1-2 times per day. In contrast, at the program level, solution providers exercise a more structured, cautious approach with releases, as they focus on coordinating and managing the integration of multiple solutions to minimize conflicts and dependencies.
- **Testing scope:** Project-based CI/CD primarily focuses on testing the functionality and behavior of a specific product. Tests are designed to validate success criteria of individual solution requirements. On the other hand, program tests require broader testing to confirm the compatibility and integration of changes. Therefore, 2030 Census Program-focused CI/CD includes a more extensive automation test suite where integration and end-to-end (E2E) tests play a more significant role.

If a third-party solution provider is performing program-level and P&S testing independently, the 2030 Census PLT team reviews the test plan, test scenarios, and appropriate test execution results to ensure the third-party test plan and execution results meet the program objectives.

2.3.1.3 Stakeholder Verification

Business stakeholders conduct tests to verify if the solution properly implements business processes and meets their defined requirements. Stakeholder verification can be grouped into five key focus areas:

- **User Integration Acceptance Testing** - Focuses on integrated, comprehensive, role-based test scenarios that interact with a solution's user interface from an E2E approach.
- **User Reports Acceptance Testing** - Focuses on validating the logic and rendering of system-generated reports to confirm report data is aggregated consistently and correctly.
- **Output Testing** - Focuses on verifying and validating the outputs, results, or deliverables of a system or process to ensure they meet expectations. This evaluation assesses the preparedness of the SoS to perform near real-time processing and post-collection processing activities.
- **Path Testing** - Focuses on testing various paths or sequences of code within a software application. Testing each point is critical to ensure no path iterates infinitely or terminates abnormally. The goal is to determine the number of unique, independent paths and uncover potential errors or logical flaws in the code by exercising different combinations.
- **Data Validation** - Data Validation testing is the process of examining the quality, accuracy, and consistency of data. Data validation tests are conducted at the original data source, throughout processing and transformations (where applicable), and after storage. This testing ensures that merged data from various sources, stored in separate repositories, are compatible and protect data from corruption.

For additional details on test efforts performed by stakeholders, refer to the TEMP, listed in Appendix C. The PRR is conducted once all functionalities of an operation have completed and passed program-level testing with no critical or major defects.

2.3.1.4 Operational Readiness Test

Operational Readiness Testing (ORT) provides stakeholders the opportunity to oversee user role-based testing that unites operational procedures and system functionality. The ORT begins after successful completion of the PRR.

ORT evaluates the SoS's readiness for production by running a series of activities that assess the 2030 Census Program's ability to function in a "real world" integrated environment and respond to a multitude of operational scenarios. Test scenarios designed for Operational Testing focus on system operations, system reliability and maintainability, process validation, and end-user training and accessibility. By conducting ORT, the program mitigates the risk of critical issues or failures occurring in a live production environment by allocating a final phase for early detection and resolution of issues.

Closeout of ORT activities kickstarts the final milestone, ORR, prior to conducting operations in Production. ORR is a final check that the needed resources (i.e., people, solutions, processes, and facilities) have been acquired, developed, and fully tested. ORR prompts solution teams to deploy code to the Production environment and complete smoke testing ahead of CO, the final milestone in which the 2030 Census Program transitions to O&M. For further details on ORT, refer to the TEMP, available in Appendix C.

2.3.1.5 Operations and Maintenance

For SoS architecture, O&M activities are crucial components that focus on verifying the ongoing functionality, performance, and support of the interconnected systems and operational procedures. The limited period of live production operations for the 2030 Census, coupled with its scale, places a demand on solution providers that often exceeds standard support models.

Operational activities for the 2030 Census Program include the following:

- **Production Checkout:** Testing that provides solution providers, Operation Delivery (OD) teams, and infrastructure teams the ability to ascertain that system configurations, authentication/connection mechanisms, and internal backend procedures are provisioned accurately and enabled prior to the full launch of production operations.
- **Soft Launch:** An early run-through of the production operation to ensure all functional and operational components, as well as data, flow through all the systems and perform as expected in advance of a full launch.
- **Monitoring:** Continuously monitoring the performance, behavior, and health of each independent solution and overall SoS. Monitoring includes collecting and analyzing data on system metrics, logs, and events to uncover anomalies or potential issues.
- **Incident Management:** Handling and resolving incidents that arise during production differs from standard pre-production defect remediation. This activity aims to standardize processes for proper identification, diagnosis, remediation, and testing of production issues while minimizing downtime and disruption.

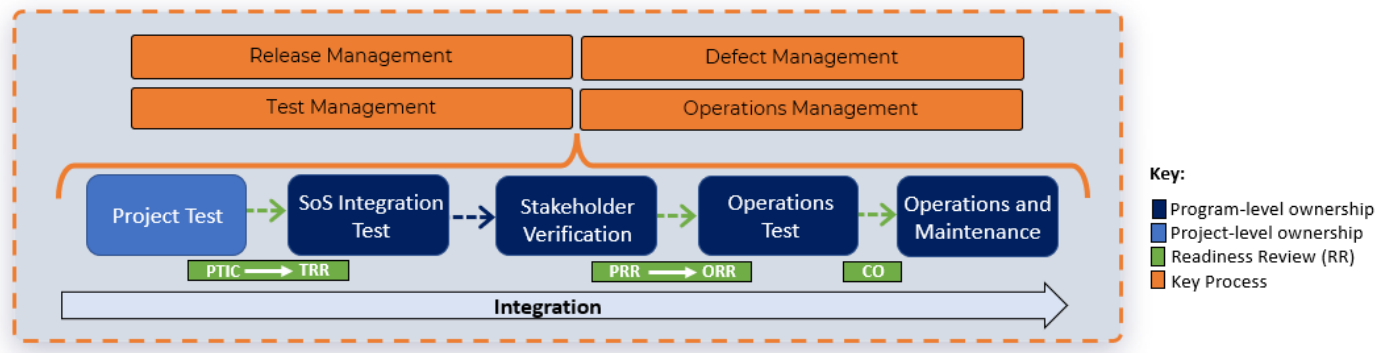
Maintenance activities include the following:

- **Preventative Maintenance:** Conducting regular maintenance activities, such as database upgrades, to prevent issues and maintain health and reliability.
- **Software Updates:** Changes to solutions are inevitable, including remediating defects identified in lower environments, promoting updated code to Production, and defects identified in Production (formally referred to as “incidents”). O&M testing is conducted for all planned and unplanned releases to Production.
- **Closeout:** Planning for the retirement or transition of a solution once its role in 2030 Census production operations concludes. When a solution reaches the end of its operational life cycle, testing of data migration/archiving and decommissioning steps are exercised.

2.3.2 Key Processes Applied - Integration

The Integration stage of the V-Model, depicted in Figure 10, is complex and necessitates close collaboration between solution providers and multiple areas of the 2030 Census Program. The following key processes applied throughout this stage are Release Management, Test Management, Defect Management, and Operations Management. These processes collectively optimize system quality, minimize downtime, enhance user experience, and align development efforts with 2030 Census Program objectives.

Figure 10: Key Processes – Integration



2.3.2.1 Release Management

The 2030 Census Program manages releases at two distinct levels: project and integration level.

Project-Level: As described in Section 2.2.1.2, solution providers are encouraged to adopt DevSecOps best practices which includes CI/CD, automation, tooling, and governance processes. The code releases may be daily or multiple times per week to the project environments.

Integration-Level: Typically beginning at PTIC, the 2030 Census Program manages deployments to the program-level environments (such as ITE and Production environments) via planned releases. These releases are typically incremental and require approval by the receiving team ahead of deployment. Approvals come from the appropriate board(s) and/or teams such as the Operations Release Board (ORB) during active production operations and the 2030 Census PLT team during program-level testing.

2.3.2.2 Test Management

2030 Census Program Test Management processes and procedures are described in the TEMP, maintained by DSE. The TEMP represents the prevailing policy, guidance, recommendations, and best practices for all testing activities that take place within the 2030 Census Program. These testing activities include the testing framework and activities planned to ensure that all aspects of the 2030 Census SoS are adequately tested and successfully implemented. The TEMP contains progressive levels of detail, starting with high-level testing guiding principles, test levels and efforts, and transitions to lower-level details including environments, roles and responsibilities, test types, and the criteria and conditions necessary for efficient and effective implementation.

The 2030 Census IT test strategy consists of four test levels: Project, Program, Operational, and Production, each with specific objectives and test scopes. At each successive level, the scope of testing becomes increasingly imitative of the Production environment. Over the course of this process, systems under test mature and become more integrated within the SoS as well as with operational and business processes. The 2030 Census Program operates using an incremental approach. By adopting an iterative methodology, the 2030 Census Program increases efficiency in the scheduling and coordination of tests efforts, strengthens flexibility, and detects defects as early as possible.

2.3.2.3 Defect Management

Detecting defects as early as possible is key to the successful implementation of the 2030 Census. Defect procedures including identification, categorization, prioritization, and reporting are outlined in the Defect Management Plan (DMP), which is maintained by the 2030 Census PLT Team.

Defect management involves documenting, prioritizing, communicating, and reporting defects found during testing of the 2030 Census systems, enabling program managers to stay informed about the status of system testing. Constant, transparent communication among 2030 Census Program leadership, stakeholders, solution providers, developers, and testers are critical to achieving successful and efficient defect remediation.

A single defect management tool is used by PLT teams to standardize how defects are logged, tracked, and resolved. This standardization offers a common framework for defect management and supports the generation of consolidated, holistic reports for all PLT efforts for the 2030 Census.

All solution providers and test teams supporting the 2030 Census Program must adhere to processes outlined with the DMP. For additional details on defect management processes and procedures, review the DMP, listed in Appendix C.

2.3.2.4 Operations Management

The IT Production Operations Management Plan (listed in Appendix C) outlines the process for managing 2030 Census production operations. The plan is of critical importance, as it centralizes O&M expectations early in the life cycle. All systems, stakeholders, and service providers should use the IT Production Operations Management Plan to plan accordingly for the effort required for production operations. In addition to the O&M processes described in The IT Production Operations Management Plan, it also includes several key components such as:

- Production Business Rhythms, such as daily production status meetings, daily defect meetings, and daily ORB meetings are implemented to ensure smooth day-to-day operations.
- 2030 Census Operations Centers and Help Desk are in place, including the Network Operations Center (NOC)/Security Operations Center (SOC), Decennial Service Desk (DSC), and War Room.
- Incident Reporting procedures and Rapid Response Team (RRT) expectations are established to provide proper communication, prioritization, and resolution of issues.

3 Systems Engineering Management

Systems Engineering Management is an ongoing process of monitoring and evaluating a program's performance to verify that it meets its goals and objectives. This verification involves measuring and analyzing schedules, RRs, and other pertinent metrics to assess program effectiveness, efficiency, and impact. DSE performance management is further complemented by consistent reporting and transparent data communication to stakeholders and decision-makers. These practices ensure that all stakeholders involved in the 2030 Census Program remain well-informed about its progress, enabling decisions to be informed by accurate and up-to-date information. The 2030 Census Program employs a multifaceted approach to measure the program's performance, encompassing various levels of management and intersecting processes as described below.

Integrated Master Schedule (IMS): The IMS incorporates business and technical schedules, including IIP milestones that align with the Work Breakdown Structure (WBS), providing a comprehensive timeline for all program activities, from planning to disposition.

IIP: The IIP outlines how IT solutions are integrated into the 2030 Census Program and confirms alignment with objectives. It also provides a high-level overview of planned ODs and associated Program RRs for ODs.

OD: The 2030 Census is a complex program consisting of multiple operations. An OD consists of a group of functionalities that are released in production at a specific time to meet operational objectives. Each OD consists of a set of systems, has defined scope, and goes through readiness milestones as defined in the IIP.

Each OD team tracks the ODs progress towards milestones and provides detailed reports to 2030 Census Program leadership. Key performance indicators (KPIs) for each milestone measure readiness for subsequent phases.

SAFe: SAFe is a methodology that provides a structured approach to align 2030 Census Program teams and all solution providers to ensure they work together toward common goals and regularly evaluate progress. SAFe employs the scaling of Agile principles and emphasizes collaboration, alignment, and delivery coordination across multiple teams working on complex programs like the 2030 Census.

As part of this methodology, Program Increments (PIs), a specified amount of time (every 8-12 weeks) with a defined purpose, allow better planning and management. Throughout the 2030 Census Program, various operations, IPTs, and OD teams use PIs to assess progress against schedules and identify potential risks.

IT Governance: IT Governance and Management establishes tools, processes, and documents to guide the 2030 Census Program, including Program Readiness Management and Technical Integration Governance.

Security Management: The solution providers work in collaboration with ISSOs and Security Engineers to provide system and application data architecture needed for determining the applicable security controls. OIS ensures compliance and grants an ATO.

Audits: Systems are subject to internal and external audits to ensure they comply with Census Bureau policies and regulations along with those of other pertinent federal agencies (i.e., Government Accountability Office (GAO)).

All IT solution providers for the 2030 Census SoS, whether internal, enterprise, or contractor-based, are expected to adhere to the 2030 Census SEMP. SEMP adherence ensures alignment

with the 2030 Census Program's objectives and standards. The DSE performance management approach provides a structured framework and guidance that navigates the 2030 Census Program toward successful execution. The following subsections expand upon each of the items noted above.

3.1 Integrated Master Schedule

The IIP framework establishes ODs and program-level review dates that are incorporated into the 2030 Census IMS. The IMS is a hierarchical and networked schedule that integrates all activities into a single timeline, including planning, design, coding, testing, deployment, and maintenance. The IMS is a comprehensive plan that includes all the sub-tasks, milestones, and deadlines needed to complete the 2030 Census.

The IMS provides a complete and accurate view of the progress of the 2030 Census and helps stakeholders identify potential risks and issues. By integrating all the project activities into a single schedule, the IMS enables teams to understand how each task impacts the overall timeline. It also verifies that all tasks are completed on time and within budget. The 2030 Census Schedule Management Plan (SMP) documents the details of the chosen schedule framework along with schedule management processes covering schedule development, the baseline, reporting, quality, risk analysis, status, and change control. The IMS complies with methodologies outlined in the SMP to make sure that a comprehensive, accurate IMS is developed with sufficient summary and detailed information to manage the 2030 Census Program.

3.2 Integration and Implementation Plan

The 2030 Census IIP describes the framework for integrating IT solutions comprising the 2030 Census architecture. By providing an actionable framework for functional integration of IT activities and operational milestones, the 2030 Census IIP helps confirm that the program's IT activities are aligned with program objectives and that IT solutions are integrated into the program's operations effectively and efficiently.

The 2030 Census IIP provides an overview of the integration framework and its two major tenets: 2030 Census Program RRs and ODs. Additional components of IIP are a list of systems that support the OD or subject-matter area, which is defined as one or a group of cross-operation functions that are managed separately and used by one or several ODs (e.g., Time and Expense functionality is used by all field ODs).

The 2030 Census IIP gives all teams the holistic view of which ODs are expected to meet which milestones when and which systems are expected to participate.

3.3 Operational Delivery Management

The Operational Delivery Management Plan (ODMP) describes how the 2030 Census Program plans, tracks, manages, and deploys ODs through their life cycle. The ODMP verifies that OD management processes adhere to the IIP schedule and procedures that focus on cross-functional coordination and iterative deliveries using SAFe guidelines. Once a determination is made regarding the number and scope of ODs, they are documented in the ODMP and represented in the IIP.

The 2030 Census Program draws on OD teams to track progress for each milestone and report to the 2030 Census leadership. An OD team consists of technical, and business leads responsible for managing the operation from development through O&M support.

OD team leads drive all aspects of the operation including establishing and managing an integrated baseline (timeline, scope, and work for the OD) and leading an interdisciplinary (such as Business, SME, Field, Logistics) team to execute the OD. The OD team leads are also:

- Responsible for driving and monitoring progress against the OD Milestone schedule. They resolve issues that impact the OD to verify the 2030 Census Program stays on track. OD team leads escalate issues to 2030 Census Program management when needed and work with other teams to determine recommendations for resolution.
- Accountable to 2030 Census Program management for requirements, development, integration/test, production readiness status, issues, risks, schedule, and cost concerns. Team leads have a deep understanding of the operations and functionality of the applications being integrated and establish the required business rhythms.

The OD team makes day-to-day decisions related to the OD and resolves issues at the lowest level possible. They also determine when a change request must be submitted against the Baseline throughout the life cycle and verify that any changes are properly documented and managed.

3.4 Scaled Agile Framework

The 2030 Census Program uses the SAFe 6.0 and leverages PIs to foster collaboration, alignment, and transparency across all solution providers and participating teams. The central aim of SAFe is to ensure a collective alignment towards shared objectives while establishing a systematic rhythm for monitoring progress and conducting evaluations at regular intervals.

SAFe 6.0 introduces an incremental process model where advancements are made through iterative checkpoints before reaching significant milestones. The iterative nature of the model empowers the 2030 Census Program to be agile, responsive, and adaptable to changes and

challenges that may arise during its execution. The SAFe allows the 2030 Census Program to operate with various methodologies including Waterfall, Agile, and Incremental that may be used across teams and parallel project efforts (e.g., contracted solutions, Chief Information Officer (CIO)). PIs and RR milestones are used to align the work priorities and assess the status regardless of the methodologies used by solution providers at key points in time.

PIs are conducted every 8-12 weeks. All teams are expected to participate in the PIs and provide commitments. PI Planning is a collaborative and structured session where solution providers come together to plan and align their work for a fixed period. Below is the high-level agenda for PI Planning:

- PI Objectives are defined, providing a clear vision of what needs to be achieved.
- Solution providers plan their work for the PI. (For example: breaking down work into user stories or tasks and document dependencies.)
- Teams identify potential risks and obstacles. Mitigation strategies are discussed with 2030 Census leadership.
- Cross-team dependencies are identified; timelines and expectations are aligned.
- Teams commit to the work they have planned for the PI.

Breaking down a program into smaller, manageable increments helps the 2030 Census Program identify and address issues early, optimize performance, and continuously improve over time. To expand collaboration with the Enterprise, the 2030 Census Program will attempt to align with Enterprise PI schedules.

At the conclusion of a PI, the 2030 Census Program assesses the progress achieved against set objectives and evaluates program performance. This evaluation encompasses metrics such as completed features or business threads, the extent of executed test cases, customer satisfaction levels, and other pertinent performance indicators, contingent upon the program's life cycle stage.

3.5 Technical Integration Governance

The 2030 Census Program recognizes the significant role IT governance boards play to optimize team performance and enable efficient, collaborative decision-making. The establishment of these boards follows a well-defined process, as outlined in the IT Governance Process document (listed in Appendix C). This process includes a precise definition of the board's scope and stakeholders, requires close coordination with existing boards and processes, establishes a cross-functional decision-making body, and provides continual assessment for improvement.

Solution providers are expected to participate in and adhere to the IT Governance process and associated boards. The specific technical boards required to govern the 2030 Census technical

integration may change as the program life cycle progresses. These technical boards are listed below, and brief descriptions are provided in Appendix D: 2030 Census IT Governance Boards.

Table 1: IT Governance Boards

Program Phase	Type of Governance Needed	Required Board(s)
Design Selection	Boards are needed to focus on supporting program research projects, requirements, and architecture as well as enabling IT innovation	Decennial Solution Review Board (DSRB) DITD EWG
Development & Integration	Project-level technical decision-making boards become critical to support DevSecOps and integration processes and manage change impacting 2030 Census systems, environments, and field tests	Technical Review Board (TRB) IT Change Approval Board (CAB)
Peak Production & Closeout	Boards are needed to protect the live environment and manage transition and closeout of operations and systems	ORB Disposition Review Board (DRB)

The interactions between related boards, especially Enterprise boards, change throughout the program life cycle. Memorandums of Agreement (MOAs) are established and consistently updated to confirm that the 2030 Census Program, its partners, and service providers are working together to plan the support required to meet program objectives. The demand placed on providers by the 2030 Census Program often exceeds any standard support model; accordingly, effective collaboration is critical for success.

3.6 Risk and Issue Management

Risk and issue management is performed using a common, systematic, and repeatable approach to verify risks, issues, and opportunities are effectively identified, managed, mitigated, and unambiguously communicated to decision-makers and stakeholders. The DITD Risk, Issue, and Opportunity Management Implementation Plan aligns to the process laid out in the 2030 Census Risk and Issue Management Plan (RIMP).

Risk, issue, and opportunity identification is not the responsibility of one individual, but the inherent responsibility of all stakeholders. Any potential risks, issues, and opportunities that may impact the success of the 2030 Census Program should be raised to management or a member of the DITD Risk and Issue Board (RIB). The identification of risks, issues, and opportunities is an iterative activity that begins during process start-up and continues regularly and throughout the program life cycle.

Once identified, the DITD RIB uses qualitative risk analysis to establish the probability rating, impact rating, risk exposure, and risk matrix. The process also prioritizes risks with potentially

high impacts in cost and/or schedule. Once completed, a risk analysis is presented to the DITD RIB for review, approval, and the assignment of owners. The DITD RIB also reviews and approves issue treatment strategies, opportunity capture strategies, and opportunity capture plans.

The goal of this process is to identify new and emerging risks, issues, and opportunities as early as possible in the life cycle. For additional information on risk and issue management activities, including mitigation and contingency planning, see the DITD Risk, Issue, and Opportunity Management Implementation Plan and RIMP, listed in Appendix C.

3.7 Change Management

Managing change is one of the biggest challenges in projects. The complexity increases for a large program, such as the 2030 Census. A formal change process uses standard procedures to efficiently manage changes, meet evolving business requirements, and minimize the impact of change.

3.7.1 2030 Census Program Change Management

Solution providers must adapt to accommodate new issues, evolving requirements, and changing business needs. The 2030 Census Program Change Management Process is in place to verify that proposed changes to the 2030 Census Program's scope, cost, and schedule are discussed, reviewed, and dispositioned. Approved changes must be properly implemented, documented, and communicated to all stakeholders. The Decennial Census Change Management Plan, listed in Appendix C, outlines the 2030 Census change request workflow and other relevant guidance.

DITD will establish a TRB to centralize management of technical integration changes impacting the 2030 Census SoS.

3.7.2 Document Management

Document management is defined as the process for receiving, reviewing, approving, and storing documents for the 2030 Census Program and project(s). These elements combine to produce consistency, quality and control, and accessibility over the 2030 Census Program and project documents.

The objectives of document management are to:

- Provide an approach to manage documents throughout their life cycle.

- Align and integrate the documents with the needs of various stakeholders and 2030 Census Program partners.
- Comply with governance requirements provided by leadership.
- Provide clear guidance for the review and approval of documents.
- Verify consistent application of the document management process across the 2030 Census Program.

Once documents are baselined and signed, any future changes must go through the formal change request process.

Document management processes are documented in the 2030 Census Document Management Plan, listed in Appendix C. Certain IT level documents are managed by DITD and are maintained in a System Engineering document repository.

4 Engineering Specialty Integration

This section covers two areas integrated across overall SEMP activities: Security Management and Assessments. These elements ensure the 2030 Census Program's compliance with guidelines, policies, and regulations. These measures demonstrate the 2030 Census Program's commitment to security, performance, and accountability, which are vital for achieving the goals of the 2030 Census Program.

4.1 Security Management

OIS staff is assigned to the DITD to provide support for the 2030 Census Program and validate data integrity, confidentiality, and availability. OIS works with the 2030 Census Program to establish programmatic standards and policy guidance to implement security requirements and comply with regulatory requirements and external guidance.

2030 Census Program solutions that require use of PII, Title 5, Title 13, or Title 26 data work with the assigned Decennial Information Security Engineer and the OIS-assigned ISSO to obtain the necessary security authorizations from the Authorizing Official (AO) (as defined by OIS Authorization Policy) for intended testing and for all systems before their release to production.

All solution providers require an ATO prior to operating in a production environment. An ATO is an official management decision given by an AO to an information system prior to the system entering operations. In accordance with the Census Bureau's IT Cybersecurity Program, there are several types of authorization methodologies used to authorize systems. Authorization methodologies currently used within the Census Bureau's IT Cybersecurity Program are Agile, Authorization to Use (ATU), Type Authorization, Cloud-Based System/Services Authorization,

and Boundary Extension Authorization. For detailed information, please refer to Authorization Policy V2.0, listed in Appendix C.

After the system goes into production, the management, operational, and technical controls are periodically assessed as part of the NIST 800-53 Continuous Monitoring phase that is required for all Census Bureau, Department of Commerce, and government systems.

The Privacy Threshold Analysis (PTA) provides a high-level description of a system and the data it manages. The Census Bureau Policy Office uses the PTA to determine whether a Privacy Impact Assessment (PIA) and/or System of Records Notice (SORN) are required. In conjunction with the OIS assigned ISSO, each solution provider completes a PTA and gains concurrence from the solution provider's Information Systems Security Manager (ITSM), then submits it to the Census Bureau privacy office for further processing. If it is determined that a PIA or SORN is required, then policies and procedures under the control of the privacy office are enacted as described on the Privacy Compliance Branch of the Policy Coordination Office intranet site.

4.2 Audits

The purpose of audits, informal or formal, is to ensure the integrity, quality, and compliance of System Engineering activities within the 2030 Census Program. Audits provide a mechanism for identifying potential issues, evaluating progress, and facilitating continuous improvement.

All technical solutions within the 2030 Census Program are subject to be verified for adherence to established guidelines and policies. Audits may be conducted by the Department of Commerce Inspector General (DOC IG), GAO, and other relevant government entities. The types of audits may include in-person interviews, in-person presentations, or submission of documents.

Appendix A: Key Stakeholders

The following table can be used as a reference for high-level ownership across DSE, TI, DCMD, and Stakeholders/Solutions Providers.

Table A-1: Key Stakeholder Responsibilities

Organization Responsible	Program Activities
Decennial Information and Technology Division (DITD)	Milestone(s): • CDR • Initial Baseline Review • TRR • Program Test Initialization Checkpoint • PRR Management: • Technical Operational Delivery • Solution Requirements • SAFe • Operations Management / O&M • IT Governance • Release Management • Technical Risks and Issues
DCMD	Milestone(s): • SRR • ORR Management: • Operational Delivery • Stakeholder Requirements • Program Risk and Issue
Solution Providers	Management: • Development and Implementation

Appendix B: Environments and Tools

Environments

The 2030 Census Program implements several different environments that align with the enterprise standards established by the Census Bureau. Whenever possible, the Census Bureau leverages a centralized set of platforms, tools, and infrastructure. The environments in Table B-1 are used at a minimum.

Table B-1: Environments

Tool	Description
Development (DEV)	This environment is used for developing and testing software. Developers conduct unit tests to make sure that the code functions properly.
Test/Quality Assurance (QA)	This environment is used for functional and non-functional testing of the code against system, business, and enterprise requirements.
ITE	This environment is used for integration and E2E testing of multiple systems. It should be as close as possible to the Production environment in terms of hardware and software.
Production (PROD)	This environment is used for live deployment of the software. It must meet capacity and availability requirements.
Staging (STG)/Pre-PROD	This environment is a pre-production environment where all E2E tests are conducted before deployment to Production. It should be like the Production environment as much as possible.

Solution Providers can deviate from these standards with their infrastructure and approved development tools if they have an environment to support the test activities.

Tools

To support System Engineering and Testing, 2030 Census Program teams use a suite of tools outlined in Table B-2 below. 2030 Census solution providers must use the tools listed within the 2030 Census IT Tool List, which align with the categories presented below.

Table B-2: Tools

Tool Category	Description
Requirements Management Tool	This tool serves as the central repository for Mission Requirements, Activity Requirements, Stakeholder Requirements, and Solution Requirements for all 2030 Census operations, providing storage and structure for requirements content and relationships.

Tool Category	Description
Defect Management Tool	The defect management tool is used to log and review defects identified during testing in all environments and phases, enabling efficient prioritization and providing defect metrics such as density, aging, and impacts.
Test Management Tool	This tool manages all testing processes for the 2030 Census Program, tracking test execution results and providing traceability of tests to requirements, as well as managing the application life cycle from requirements through testing and defect reporting.
Model-Based Systems Engineering (MBSE) Tool	This architecture tool is used to build architecture diagrams including application diagrams, WFDs, and model business processes for each of the ODs in the 2030 Census Program. The architecture tool presents the relationships between system components and interface paths.
BPM Tool	A BPM visualization tool is used to build business process models that depict the requirement capabilities as they would be implemented in the 2030 Census business implementation.
Helpdesk Ticket Management Tool	A helpdesk management system is used to track the resolutions/fixes applied to the system software to address defects identified during production operations.
Interface Management Tool	A collaborative API management tool is used to generate standardized reference material for all solution providers, promoting component reuse, standard formatting, and information sharing.
Schedule Management Tool	The schedule management tool is used to construct schedules of work for all activities on the 2030 Census Program. This schedule enables critical path assessment and impact evaluation on downstream systems caused by delays from upstream systems.
Program Management Tool	This tool is designed to help solution providers and program managers plan, execute, and track multiple projects, or initiatives, under a large program. This tool is a centralized platform that manages various aspects of a program including planning, progress tracking, collaboration, and reporting to provide a holistic view of program status.
DevSecOps Pipeline Tool	A software tool that automates the CI/CD processes for software development. These tools help development teams build, test, and deploy code changes rapidly and reliably.

Tool Category	Description
Security Scanning Tool(s)	This tool, or tools, identify and assess security vulnerabilities and weaknesses in software applications, and therefore play(s) a crucial role in ensuring the security and integrity of applications. The tool supports various scanning, including: • Code Quality • Secrets Detection • Static Application Security Testing (SAST) • Dependency Scanning • IaC Scanning • Container Scanning • Dynamic Application Security Testing (DAST) • Interactive Application Security Testing (IAST) • Fuzz Testing
Monitoring Tool	This tool observes, tracks, and collects data from systems, networks, and infrastructure components to provide real-time insights into the health, performance, and availability of applications.
Risk Management Tool	A tool used to facilitate the identification, assessment, tracking, and mitigating of risks. This tool helps solution providers proactively manage risk, make informed decisions, and take appropriate actions to minimize impact.
Cost/Budget Tool	This tool is designed to assist organizations in planning, managing, and tracking financial resources and expenses. Features of the tool should include the ability to create budgets, monitor spending, analyze financial data, and build financial reports that offer insights into spending patterns and overall financial health.
Financial Operations Tool	Software to understand cloud consumption, translate complex billing and invoices, correlate usage and costs against users or projects, maintain reports and alerts, and assist in the decisions that can optimize cloud usage and costs.

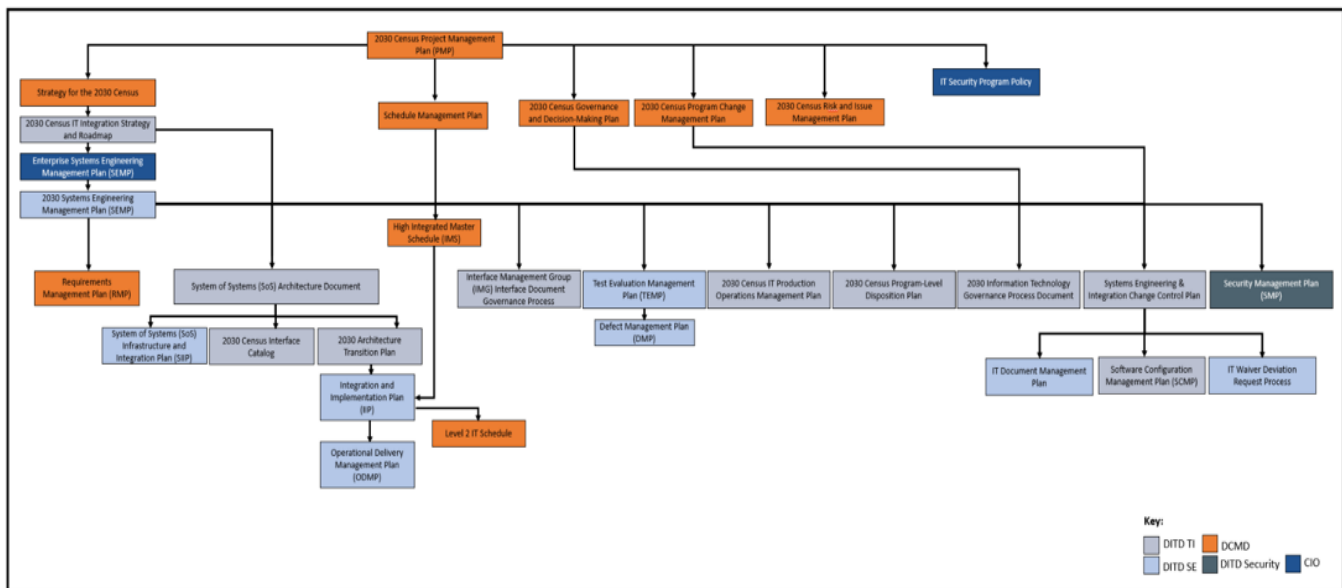
Appendix C: Referenced Documents

DITD maintains the *Planning and Coordination Documents for the 2030 Census Program*, which is an active directory for the 2030 Census Program. (Examples include the DMP, ODMP, the IIP, and others.)

Items on this list are subject to change based on the needs of the 2030 Census Program. As new versions of each document become available, the links in this index are updated to the latest complete version.

Figure C-1 below is a representation of the relationships between the various planning and coordination documents for the 2030 Census Program.

Figure C-1: 2030 Census Document Relationships



Appendix D: 2030 Census IT Governance Boards

Table D-1: 2030 Census IT Governance Boards

IT Governance Board	Description
2030 Census Decennial Solution Review Board (DSRB)	This DCMD-owned board provides decision-making authority for operational and IT stakeholder requirements, recommendations on build, buy, or reuse solution decisions, and oversight of Decennial Program Architecture.
DITD Engineering Working Group (EWG)	This DITD-owned board has two specific scope areas: First, it manages DITD and 2030 Census Analysis of Alternatives (AoAs). Second, it provides support and governance to the Decennial Census Programs Directorate and 2030 Census systems as they plan for technological transformation, with a focus on cloud smart architecture.
2030 Census Technical Review Board (TRB)	This DITD-owned board centralizes management of technical integration changes impacting the 2030 Census SoS.
DITD IT Change Approval Board (CAB)	This DITD-owned board assesses infrastructure changes to the 2030 Census infrastructure.
2030 Census Operational Release Board (ORB)	This DITD-owned board centralizes approval of deployments to the production environment during 2030 Census IT Production Operations.
2030 Census Disposition Review Board (DRB)	This DITD-owned board centralizes approval for systems reductions/decommissions/migration plans, as well as infrastructure builds and timelines for 2030 Census closeout.